Agentic AI System Design Report

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Overview

This project implements a semi-autonomous clinical operations triage agent. The system goal is to route structured review cases to standard queue, data-quality review, expedited human review, or mandatory human approval. It is deliberately bounded: the agent does not diagnose patients, recommend treatment, or make final patient-facing decisions.

System Architecture and Design

The agent includes a persona and scope prompt, a policy configuration file, a reasoning loop, persistent JSON memory, and three tools. The data-quality tool counts missing fields and estimates quality risk. The policy lookup tool checks whether the case is patient-facing. The priority tool combines model risk, confidence gaps, and data-quality risk into a routing score. This mirrors the reasoning-and-acting pattern in agentic systems, where a system alternates between reasoning steps and tool observations (Yao et al., 2023).

Decision Logic and Behavior

The demo executed 4 representative cases. The observed decision counts were {'standard_queue': 1, 'expedited_human_review': 1, 'human_review_required': 1, 'manual_data_quality_review': 1}. A low-risk internal case entered the standard queue, a case with missing operational fields was escalated, a patient-facing case required human approval, and an invalid case triggered fallback because a required field was absent. The trace field in each output records initialization, tool calls, observations, and final reasoning.

Safety, Reliability, and Transparency

The most important safeguard is a hard boundary that patient-facing cases always require human approval. A second safeguard is schema validation: if required fields are missing, the agent stops normal routing and sends the case to manual data-quality review. These controls align with AI risk-management guidance that emphasizes transparency, accountability, and human oversight for consequential systems (NIST, 2023).

Observed Behavior and Limitations

The workflow behaved as designed on the four demo cases and saved both execution logs and memory. Its limitation is that the risk score is a rule-based operational proxy rather than a validated clinical measure. The agent also depends on structured inputs; unstructured notes, ambiguous cases, and adversarial inputs would require stronger parsing, validation, and testing.

Ethical and Responsible Use Considerations

The main ethical concern is inappropriate automation in a healthcare context. Even a routing assistant can create harm if it delays urgent cases, hides uncertainty, or appears more authoritative than it is. The design therefore keeps decisions transparent, logs tool observations, and requires human approval for patient-facing outcomes.

Future Improvements

Future work should add unit tests, richer scenario coverage, confidence calibration, role-based access controls, red-team cases, and monitoring dashboards. A production version should include human feedback loops and regular audits of escalation patterns across patient groups.

References

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